Design Problem Trade Studies

(Rev-S S26)

EEE 5557 Intro to Radar

Where to start?

First step:

- You should determine which are your design parameters?
  - These are the “things” that you can adjust

Design Parameters
(things you can adjust to meet the Perf Spec)

- UAV altitude
  - **Predator UAV Operations:**
  - **UAV: altitude**: 2.5 – 7.5 km (in steps of 0.5 km) ??? (straight and level flight);

You are free to pick any altitude for your design, but for the design competition, you must justify why you chose a particular value (trade study)

Design Parameters
(things you can adjust to meet the Perf Spec)

- UAV altitude
  - **Predator UAV Operations:**
  - **UAV: altitude**: 2.5 – 7.5 km (in steps of 0.5 km) ??? (straight and level flight);

You are free to pick any altitude for your design, but for the design competition, you must justify why you chose a particular value (trade study)

- Higher altitude will increase the range for a selected grazing angle
  - Improves: warning time, swath width, antenna footprint length (reduces spin rate for “N” scans & thereby increases POT)

Design Parameters
(things you can adjust to meet the Perf Spec)

- UAV altitude
  - **Predator UAV Operations:**
  - **UAV: altitude**: 2.5 – 7.5 km (in steps of 0.5 km) ??? (straight and level flight);

You are free to pick any altitude for your design, but for the design competition, you must justify why you chose a particular value (trade study)

- Higher altitude will increase the range for a selected grazing angle
  - Improves: warning time, swath width, antenna footprint length (reduces spin rate for “N” scans & thereby increases POT)
  - Degrades: PRF for one pulse on surface, requires larger antenna for gain &/or higher Xmt power for Pr (to compensate for increased range), increases the pulse IFOV (more clutter cross section requires a shorter Xmt pulse to reduce del-R, which increases the noise power)

Design Parameters
(things you can adjust to meet the Perf Spec)

- Radar freq
  - Transmitter:
    - Freq: ??? either: 37 GHz, 90 GHz or 150 GHz;

You are free to pick any altitude for your design, but for the design competition, you must justify why you chose a particular value (trade study)

Design Parameters
(things you can adjust to meet the Perf Spec)

- Radar freq
  - Transmitter:
    - Freq: ??? either: 37 GHz, 90 GHz or 150 GHz;

You are free to pick any altitude for your design, but for the design competition, you must justify why you chose a particular value (trade study)

- Higher freq ”make the antenna larger”
  - Improves: narrower beamwidth = higher gain, narrow Az beamwidth reduces clutter , narrow El beamwidth raises available PRF (single pulse on surface)
  - Degrades: higher atmos atten – requires higher Xmt power

Design Parameters
(things you can adjust to meet the Perf Spec)

- Antenna boresight pointing (grazing angle)

  Boresight: Elevation boresight grazing angle ψ_{bore} = ???? ; but $\psi_{far} \geq 2^\circ$

Major trade parameter
You are free to pick any grazing angle for your design, but you must justify why you chose a particular value (trade study)

Design Parameters
(things you can adjust to meet the Perf Spec)

- Antenna boresight pointing (grazing angle)

  Boresight: Elevation boresight grazing angle ψ_{bore} = ???? ; but $\psi_{far} \geq 2^\circ$

  Major trade parameter
  You are free to pick any grazing angle for your design, but you must justify why you chose a particular value (trade study)

  Target & clutter cross sections vary with grazing ang
  - Pick optimum grazing ang to increase T/C
  - Grazing ang & altitude are coupled - both should be considered together

  Small grazing ang puts the SNR in a T/N region
  Large grazing ang puts the SNR in a T/C region

Design Parameters
(things you can adjust to meet the Perf Spec)

- Antenna elevation & azimuth beamwidth (Ant Diameter)

Coupled with the Alt and grazing ang trade study

Also, a design goal to reduce antenna size

Design Parameters
(things you can adjust to meet the Perf Spec)

- Antenna spin rate
- Very important for number of scans target tracking
  - Higher spin rates
    - Improves: # spins tracking
    - Degrades: TOT and POT
  - Lower spin rates
    - Improves: TOT & POT
    - Satisfies a design goal

Design Parameters
(things you can adjust to meet the Perf Spec)

- Transmit power & pulse width
- Transmit power increases both T & C but not affect N
  - Improves: T/N SNR increases Pd
  - Degrade: design goal for reduced Xmt power

Design Parameters
(things you can adjust to meet the Perf Spec)

- Transmit power & pulse width
- Transmit power increases both T & C but not affect N
  - Improves: T/N SNR increases Pd
  - Degrade: design goal for reduced Xmt power
- Pulse width decrease
  - Improves: reduces del-R clutter cross section, requirement to make del-R > 6 m
  - Degrades: Increases bandwidth (more noise power), which degrades T/N ratio

Design Parameters
(things you can adjust to meet the Perf Spec)

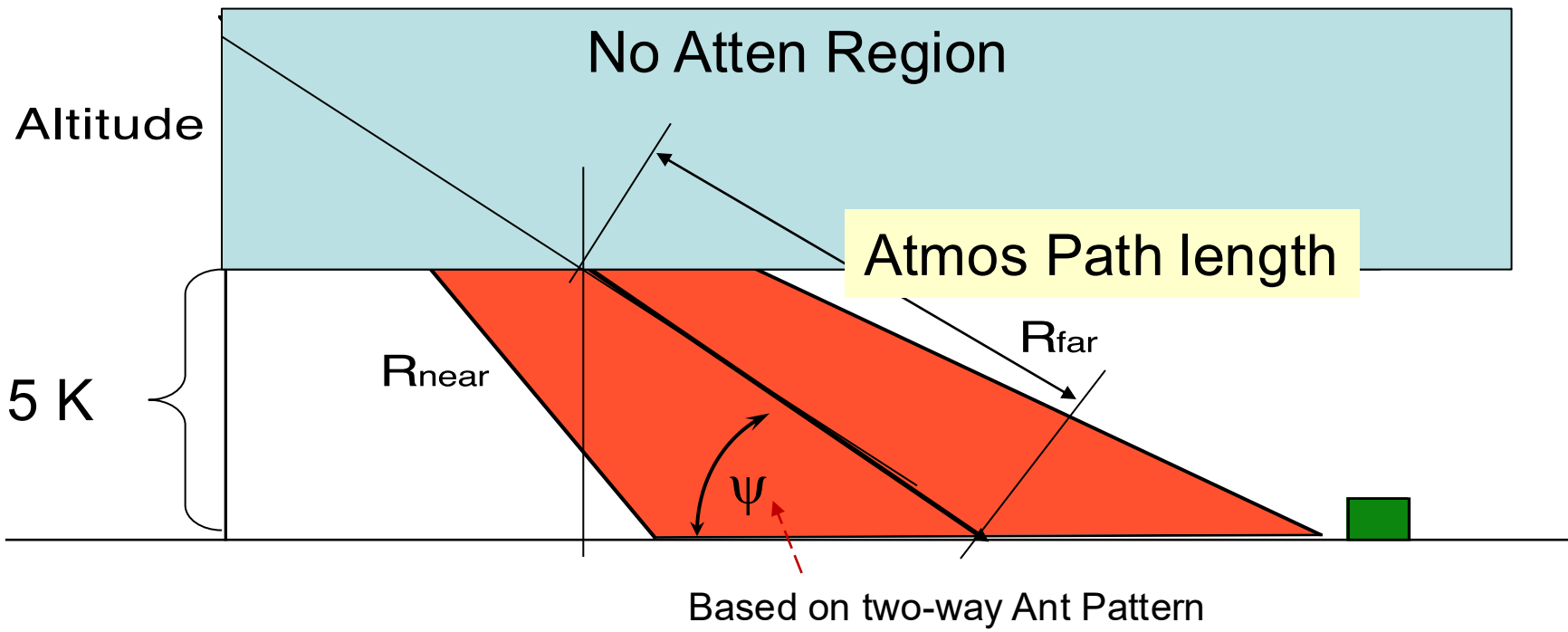
- PRF
- More is better to increase PRF, which improves Pd
  - Limited by single pulse on surface, which is delta-R (Rfar – Rnear) – controlled by $Beta_{EL}$ and grazing ang
- However, for POT > 100 there is significantly reduced benefit – lower $E_i(n)$

Design Parameters
(things you can adjust to meet the Perf Spec)

- <Tfa>
- Sets the receiver threshold (lower is better) & P_{fa}; however, requirement ≥ 10 * (Ant scan period)
- <Tfa > = 1/ (P_{fa} * B) where B = 1/ tau

- note that this is independent of cum P(2 of 3)!

Calc Atmos Path Loss for Alt ≥ 5.0 km



Max Atmos Path length = 5.0 / sin(ψ)

Must calc for near, bore & far ranges

Path Loss = 2* path_length (km) * Atmos absorption coeff

Design Parameters
(things you can adjust to meet the Perf Spec)

- UAV altitude
- Radar freq
- Antenna boresight pointing (grazing angle)
- Antenna elevation & azimuth beamwidth
- Antenna spin rate
- Transmit power & pulse width
- PRF
- <Tfa>
- Simultaneous Target detection & Doppler or Sequential Target detection & Doppler
- Pulse Compression
- Other?

Second step: Design Constraints

- You should determine your design constraints?

Second step: Design Constraints

- You should determine your design constraints?
  - UAV altitude range (max/min)

Second step: Design Constraints

- You should determine your design constraints?
  - UAV altitude range (max/min)
  - Max RPM

Second step: Design Constraints

- You should determine your design constraints?
  - UAV altitude range (max/min)
  - Max RPM
  - Max antenna dimensions (az & el)

Second step: Design Constraints

- You should determine your design constraints?
  - UAV altitude range (max/min)
  - Max RPM
  - Max antenna dimensions (az & el)
  - Min grazing angle (range-far)

Second step: Design Constraints

- You should determine your design constraints?
  - UAV altitude range (max/min)
  - Max RPM
  - Max antenna dimensions (az & el)
  - Min grazing angle (range-far)
  - PRF max (single pulse on surface)

Second step: Design Constraints

- You should determine your design constraints?
  - UAV altitude range (max/min)
  - Max RPM
  - Max antenna dimensions (az & el)
  - Min grazing angle (range-far)
  - PRF max (single pulse on surface)
  - PRF min (nyquist sampling for Doppler)

Second step: Design Constraints

- You should determine your design constraints?
  - UAV altitude range (max/min)
  - Max RPM
  - Max antenna dimensions (az & el)
  - Min grazing angle (range-far)
  - PRF max (single pulse on surface)
  - PRF min (nyquist sampling for Doppler)
  - Min pulse resolution (az & range)

Based on two-way
Ant Pattern

Second step: Design Constraints

- You should determine your design constraints?
  - UAV altitude range (max/min)
  - Max RPM
  - Max antenna dimensions (az & el)
  - Min grazing angle (range-far)
  - PRF max (single pulse on surface)
  - PRF min (nyquist sampling for Doppler)
  - Min pulse resolution (az & range)
  - Max Xmt power & duty cycle

Second step: Design Constraints

- You should determine your design constraints?
  - UAV altitude range (max/min)
  - Max RPM
  - Max antenna dimensions (az & el)
  - Min grazing angle (range-far)
  - PRF max (single pulse on surface)
  - PRF min (nyquist sampling for Doppler)
  - Min pulse resolution (az & range)
  - Max Xmt power & duty cycle
  - Others?

Third Step: Perf Specs & Goals

- You should determine your performance specs?

Third Step: Perf Specs & Goals

- You should determine your performance specs?
  - Swath width (spec & goal)

Third Step: Perf Specs & Goals

- You should determine your performance specs?
  - Swath width (spec & goal)
  - Warning time (spec & goal)

Third Step: Perf Specs & Goals

- You should determine your performance specs?
  - Swath width (spec & goal)
  - Warning time (spec & goal)
  - Prob of detection (spec & goal): Pd(2 of 3)

Third Step: Perf Specs & Goals

- You should determine your performance specs?
  - Swath width (spec & goal)
  - Warning time (spec & goal)
  - Prob of detection (spec & goal)
  - Target detection for consecutive scans (spec & goal)

Third Step: Perf Specs & Goals

- You should determine your performance specs?
  - Swath width (spec & goal)
  - Warning time (spec & goal)
  - Prob of detection (spec & goal)
  - Target detection for consecutive scans (spec & goal)
  - Doppler meas precision (spec & goal)

Third Step: Perf Specs & Goals

- You should determine your performance specs?
  - Swath width (spec & goal)
  - Warning time (spec & goal)
  - Prob of detection (spec & goal)
  - Target detection for consecutive scans (spec & goal)
  - Doppler meas precision (spec & goal)
  - Minimum <Tfa>

Third Step: Perf Specs & Goals

- You should determine your performance specs?
  - Swath width (spec & goal)
  - Warning time (spec & goal)
  - Prob of detection (spec & goal)
  - Target detection for consecutive scans (spec & goal)
  - Doppler meas precision (spec & goal)
  - Minimum <Tfa>
  - Minimum Transmitter power (goal)

Third Step: Perf Specs & Goals

- You should determine your performance specs?
  - Swath width (spec & goal)
  - Warning time (spec & goal)
  - Prob of detection (spec & goal)
  - Target detection for consecutive scans (spec & goal)
  - Doppler meas precision (spec & goal)
  - Minimum <Tfa>
  - Minimum Transmitter power (goal)
  - Others?

Design Trades Example

The following design trades are presented in an arbitrary order – NOT optimized order. You should think about your trades before starting.

Example Min Warning Time Req’mt Design Trades

Example Min Warning Time Reqmt Design Trades

- Given a min warning time req'mt
  - **Minimum Target Detection Time (warning time) T_{war}:** Using the worst-case target velocity (speed and direction of travel) and swath location, the **requirement** is:
    - to detect the target **at least 6 sec** before the aircraft reaches the target (i.e., the point of closest approach is when the target crosses the cross-track plane)

Min Warning Time Reqmt Design Trades

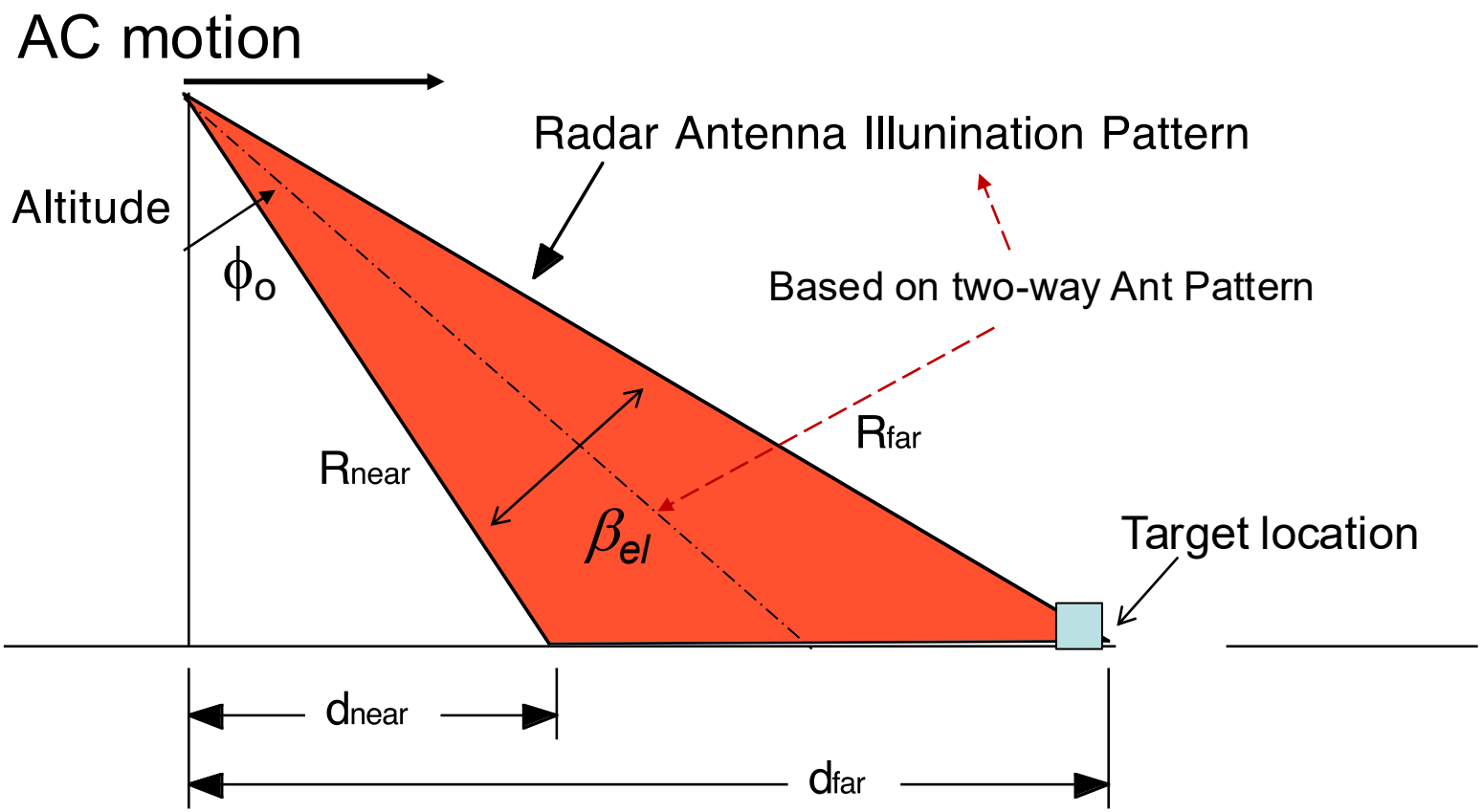
- Given a min warning time reqmt
  - **Minimum Target Detection Time (warning time) T_{war}:** Using the worst-case target velocity (speed and direction of travel) and swath location, the **requirement** is:
    - to detect the target **at least 6 sec** before the aircraft reaches the target (i.e., the point of closest approach is when the target crosses the cross-track plane)
    - This requirement allows time for a camera with cross-track pointing capability to be pointed at the target as the aircraft flies overhead (or passes by)

Min Warning Time Reqmt Design Trades

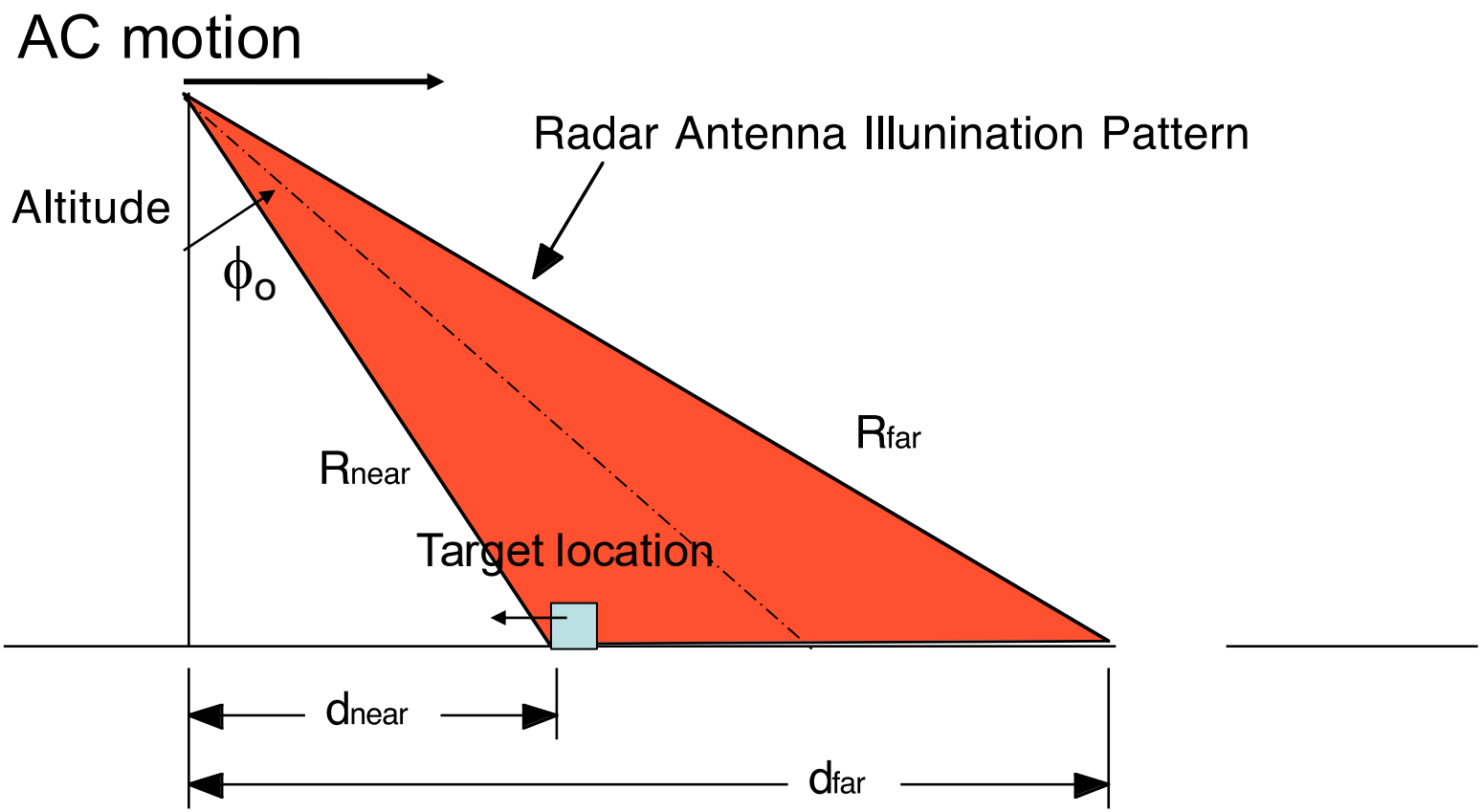
- Given a min warning time reqmt
  - **Minimum Target Detection Time (warning time) T_{war}:** Using the worst-case target velocity (speed and direction of travel) and swath location, the **requirement** is:
    - to detect the target **at least 3 sec** before the aircraft reaches the target (the point of closest approach is when the target crosses the cross-track plane)
    - This requirement allows time for a camera with cross-track pointing capability to be pointed at the target as the aircraft flies overhead (or passes by)
    - The **goal** is to increase warning time up to **60 sec**. Bonus evaluation points will be assigned for this enhanced capability
  - What parameters affect the T_{war}?

Antenna Footprint & Target Location for Best Case (Max Warning Time),

(Az = 0° and Target directly ahead of UAV)

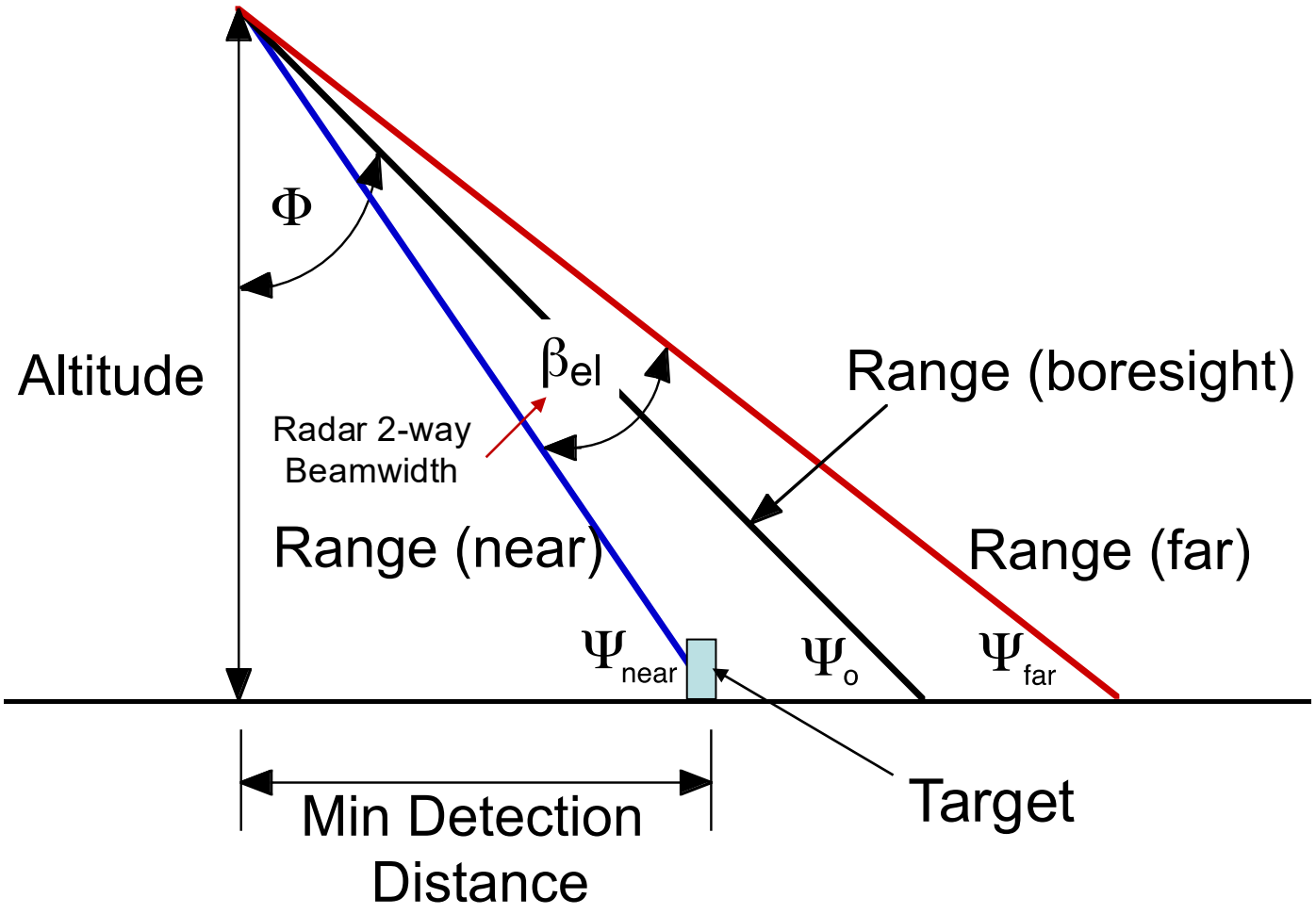


Antenna Footprint & Target Location for Worst Case (Min Warning Time)



Minimum Detection Distance Calc

Definition: Closest distance that a target may be before detection



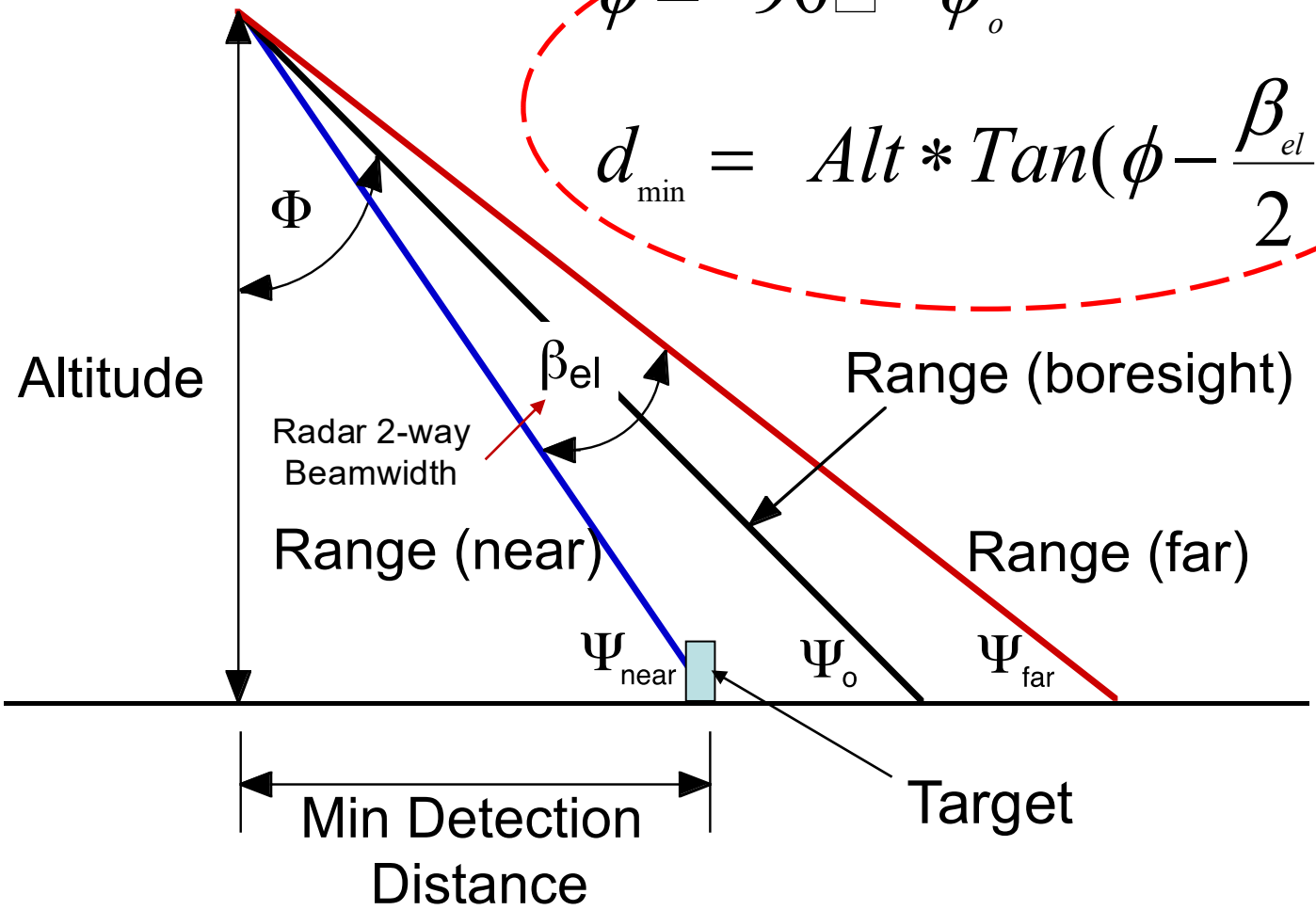
Minimum Detection Distance Calc

Definition: Closest distance that a target may be before detection

Boresight antenna pointing

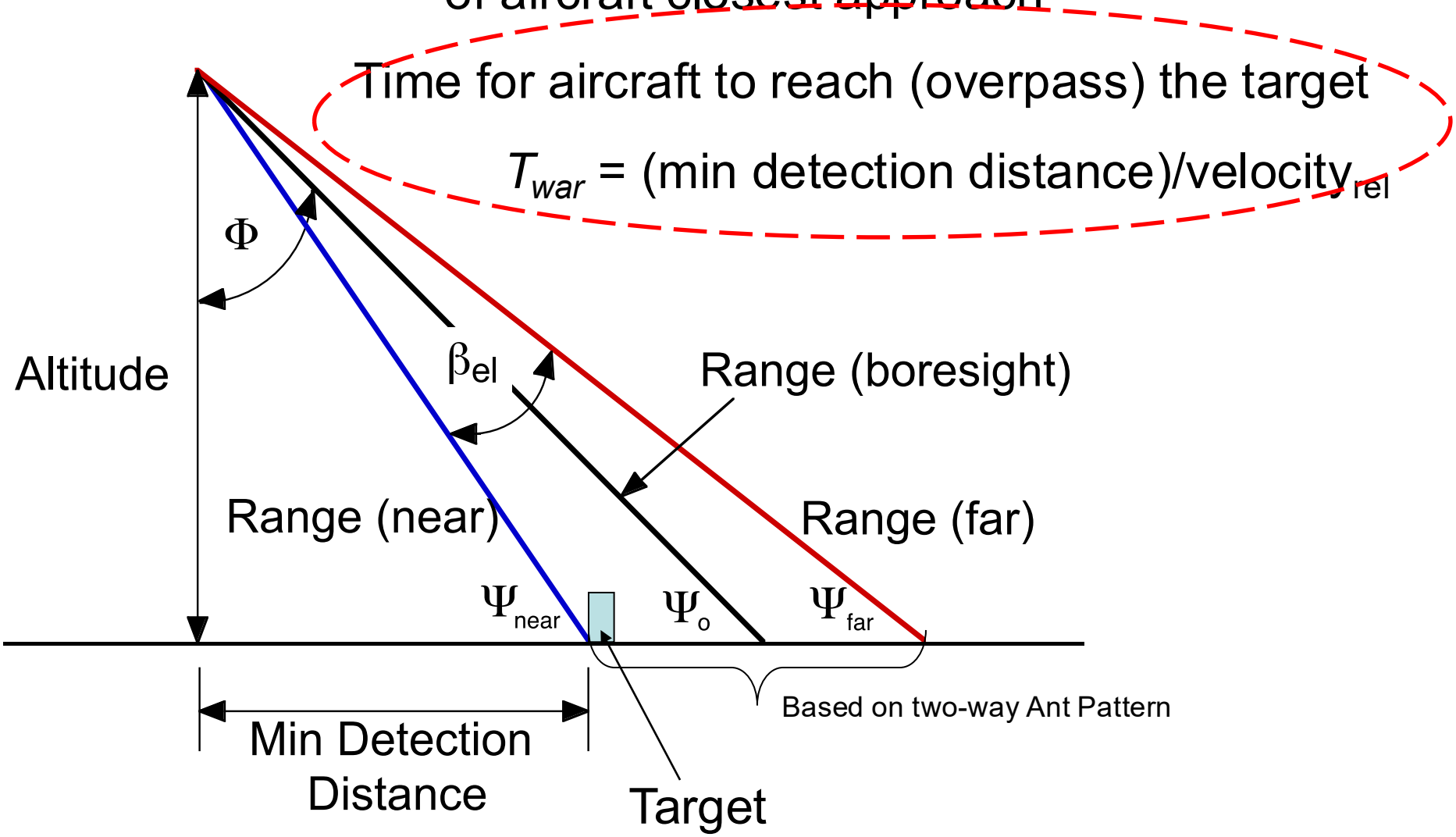
$$\phi = 90 - \psi_o$$

$$d_{min} = Alt * Tan(\phi - \frac{\beta_{el}}{2})$$



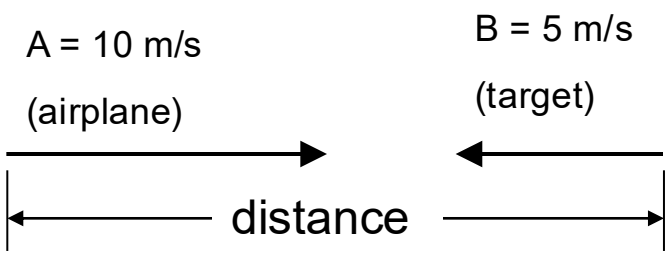
Minimum Warning Time (T_{war})

Definition: time delta between first target detection and the time of aircraft closest approach



Target Relative Velocity along UAV Track

Calculate Relative Velocity



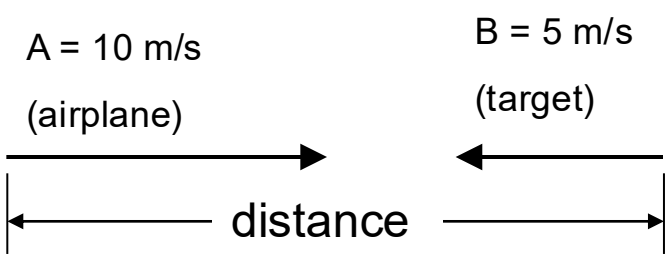
What is the relative velocity

of B with respect to A?

10 m/s – (5 m/s*cos(180°)) = 15 m/s

Target Relative Velocity along UAV Track

Calculate Relative Velocity



What is the relative velocity

of B with respect to A?

10 m/s – (5 m/s*cos(180°)) = 15 m/s

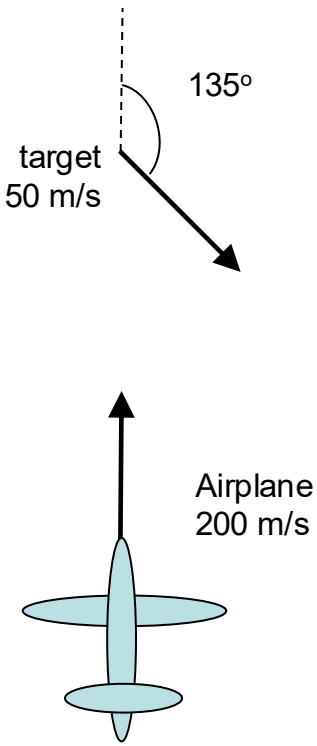
What is the relative velocity

of B with respect to A?

10 m/s – (5 m/s*cos(0°)) = 5 m/s

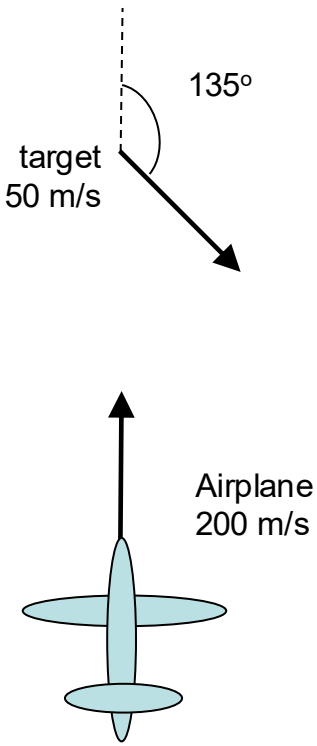
Minimum Warning Time Calc cont.

Airplane flying at an altitude of 5000 meters with a velocity of 200 m/s due North, the bore-grazing angle is 25°, Elevation HPBW is 10°. If a target is moving SSE (135° from North) at 50 m/s find the minimum warning time.



Minimum Warning Time Calc cont.

Airplane flying at an altitude of 5000 meters with a velocity of 200 m/s due North, the bore-grazing angle is 25°, Elevation HPBW is 10°. If a target is moving SSE (135° from North) at 50 m/s find the minimum warning time.

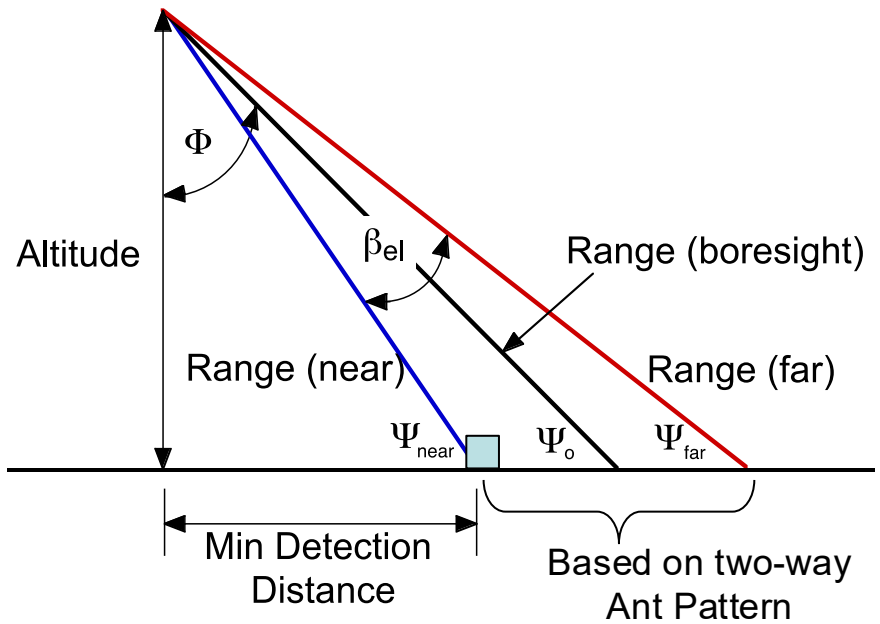


Calc. min distance (target @ R_{near}):

Cone ang @ Rnear =

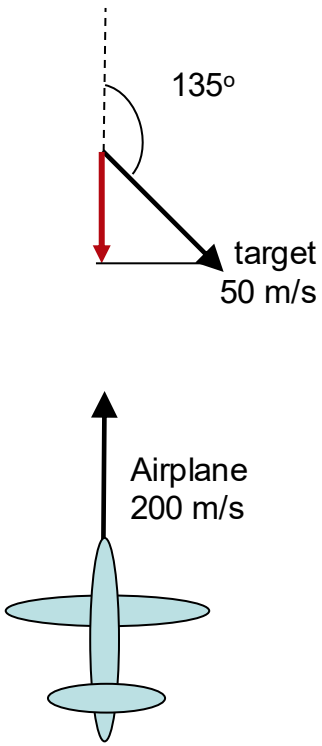
$$\phi - \beta_{el}/2 = (90^o - 25^o) - \left(10^o/2\right) = 60^o$$

$$Dist_{\min-\det} = 5000m * \tan(60^o) = 8660.3 meters$$



Minimum Warning Time Calc cont.

Airplane flying at an altitude of 5000 meters with a velocity of 200 m/s due North, the grazing angle is 25°, Elevation HPBW is 10°. If a target is moving SSE (135° from North) at 50 m/s find the minimum warning time.

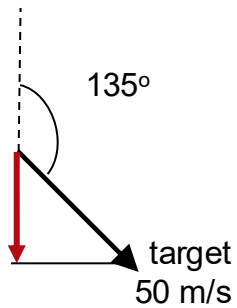


Calc relative velocity:

$$V_{rel} = 200m/s - 50m/s * \cos(135^o) = 235.4m/s$$

Minimum Warning Time Calc cont.

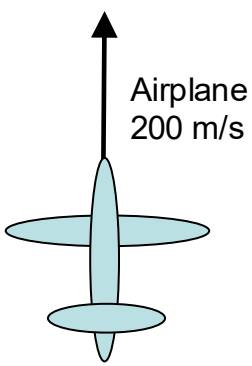
Airplane flying at an altitude of 5000 meters with a velocity of 200 m/s due North, the grazing angle is 25°, Elevation HPBW is 10°. If a target is moving SSE (135° from North) at 50 m/s find the minimum warning time.



Calc relative velocity:

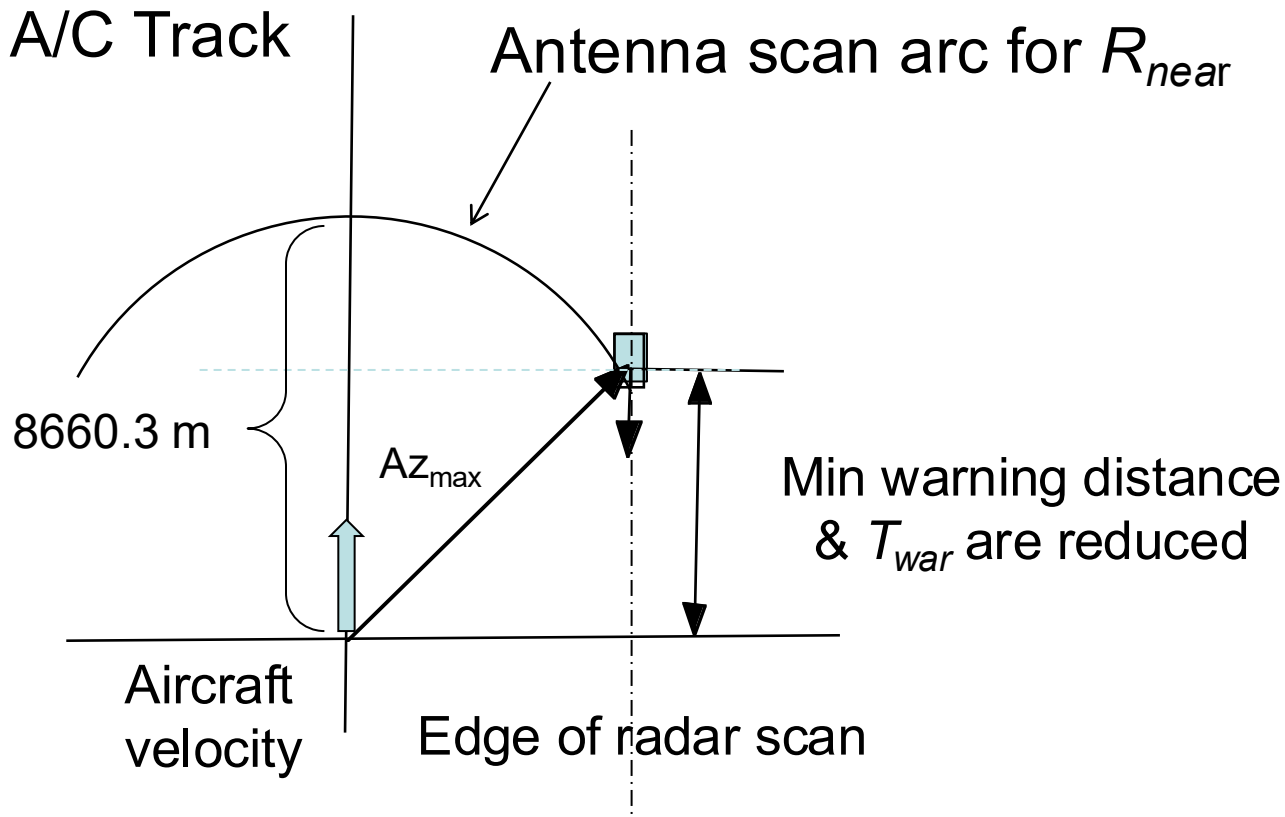
$$V_{rel} = 200m/s - 50m/s * \cos(135^o) = 235.4m/s$$

Calc min warning time:



$$t_{\min} = \frac{8660.3m}{235.4m/s} = 36.8\sec$$

Minimum Warning Time for Target Occurs @ Edge of Scan



$$T_{war_min} = T_{war_along\text{-}trk} * \cos(Az_{max})$$

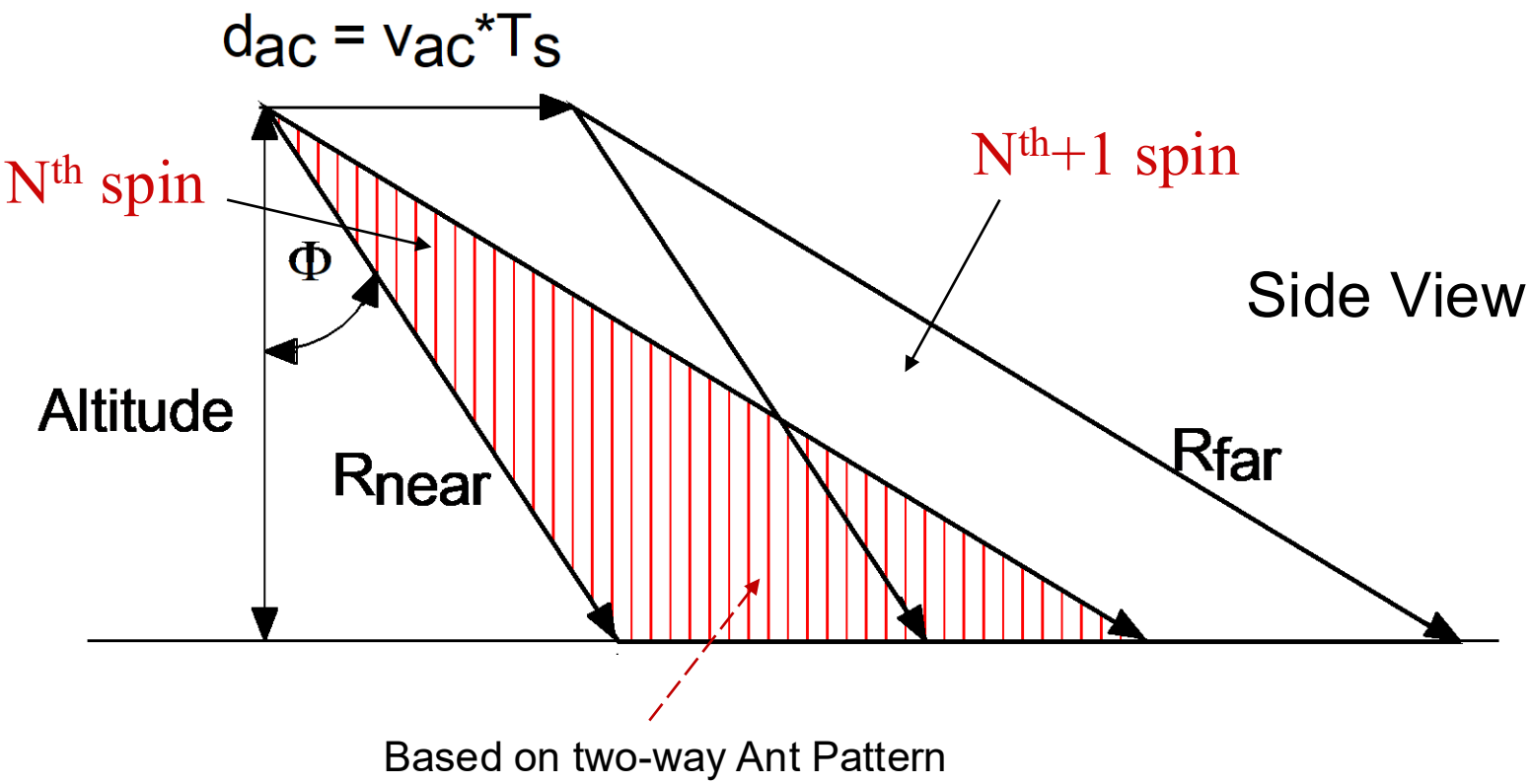
Example Minimum Spin Rate

Example Minimum Spin Rate

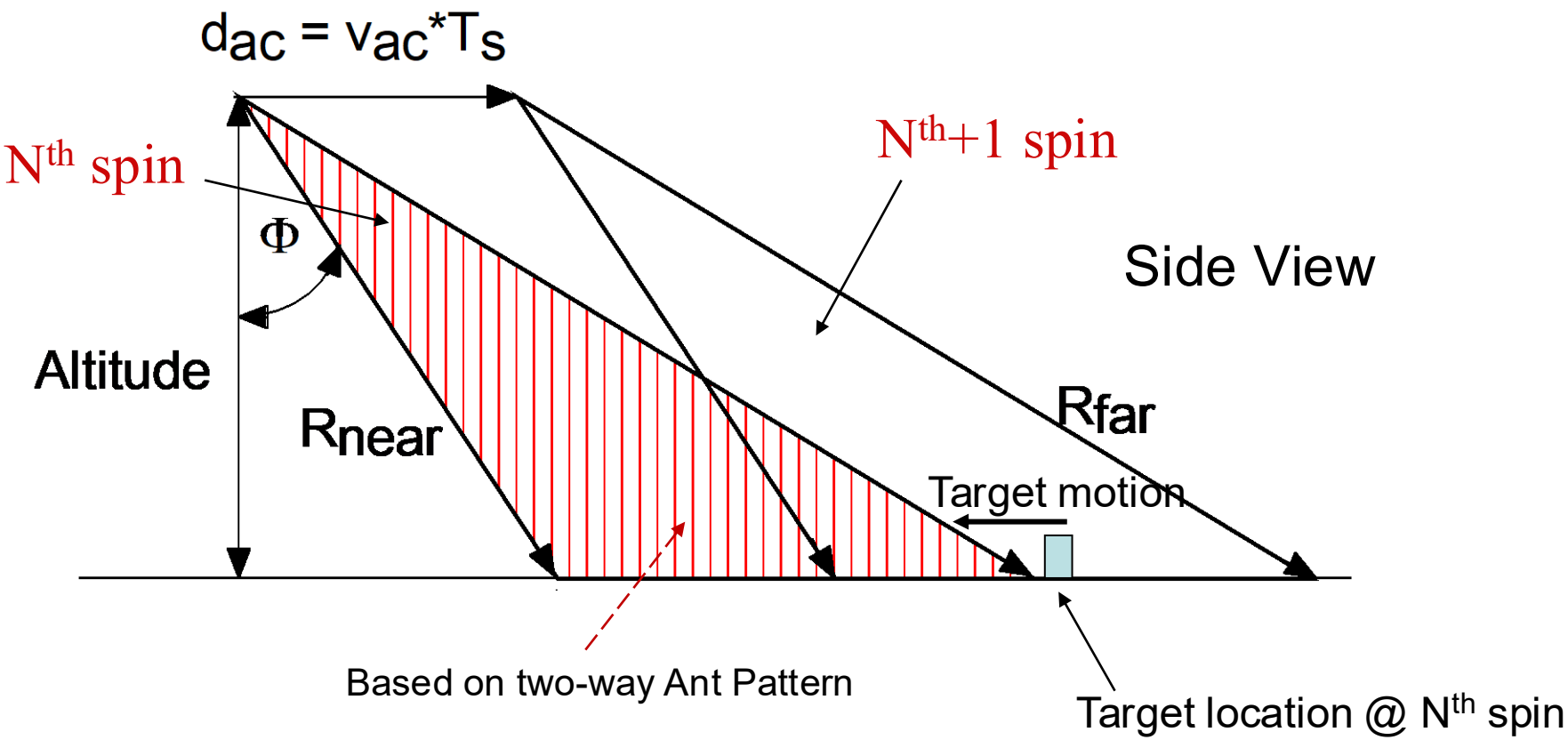
- Given a min warning time reqmt
  - You have selected an antenna boresight Ψ_o and the β_{EL} that satisfies T_{war} worst-case
  - Now for this antenna pointing configuration, calculate the minimum spin rate to ensure <u>not missing a target</u>

Minimum Antenna Footprint Overlap for two consecutive scans – to prevent missing Target

- Consider the spinning antenna footprints for consecutive scans (spin #'s)

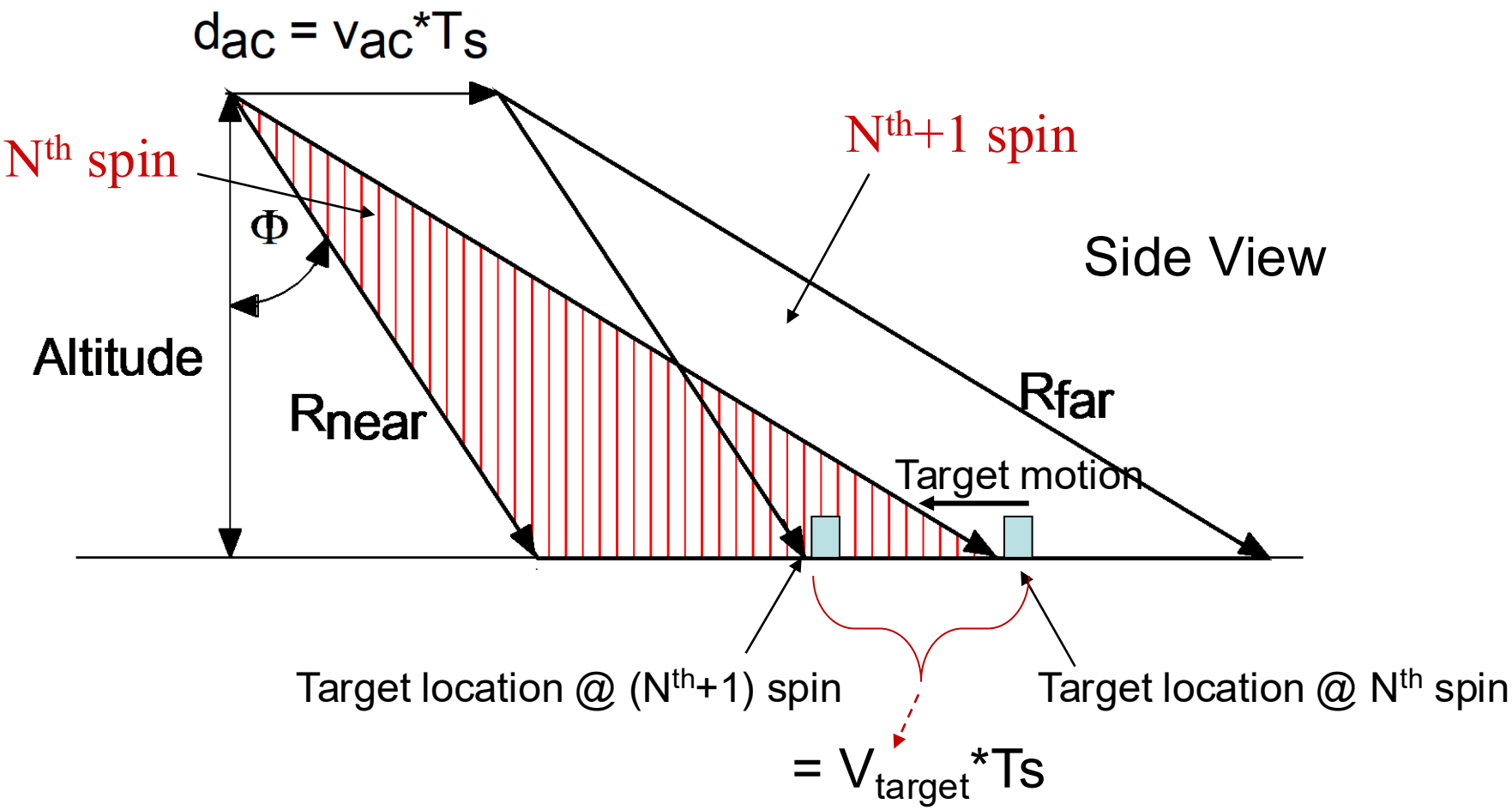


- Consider a target located "just outside" of antenna foot print in the Nth spin
  - Antenna must spin rapidly enough to ensure that a moving target will be observed
    - This requires that the footprints overlap between scans

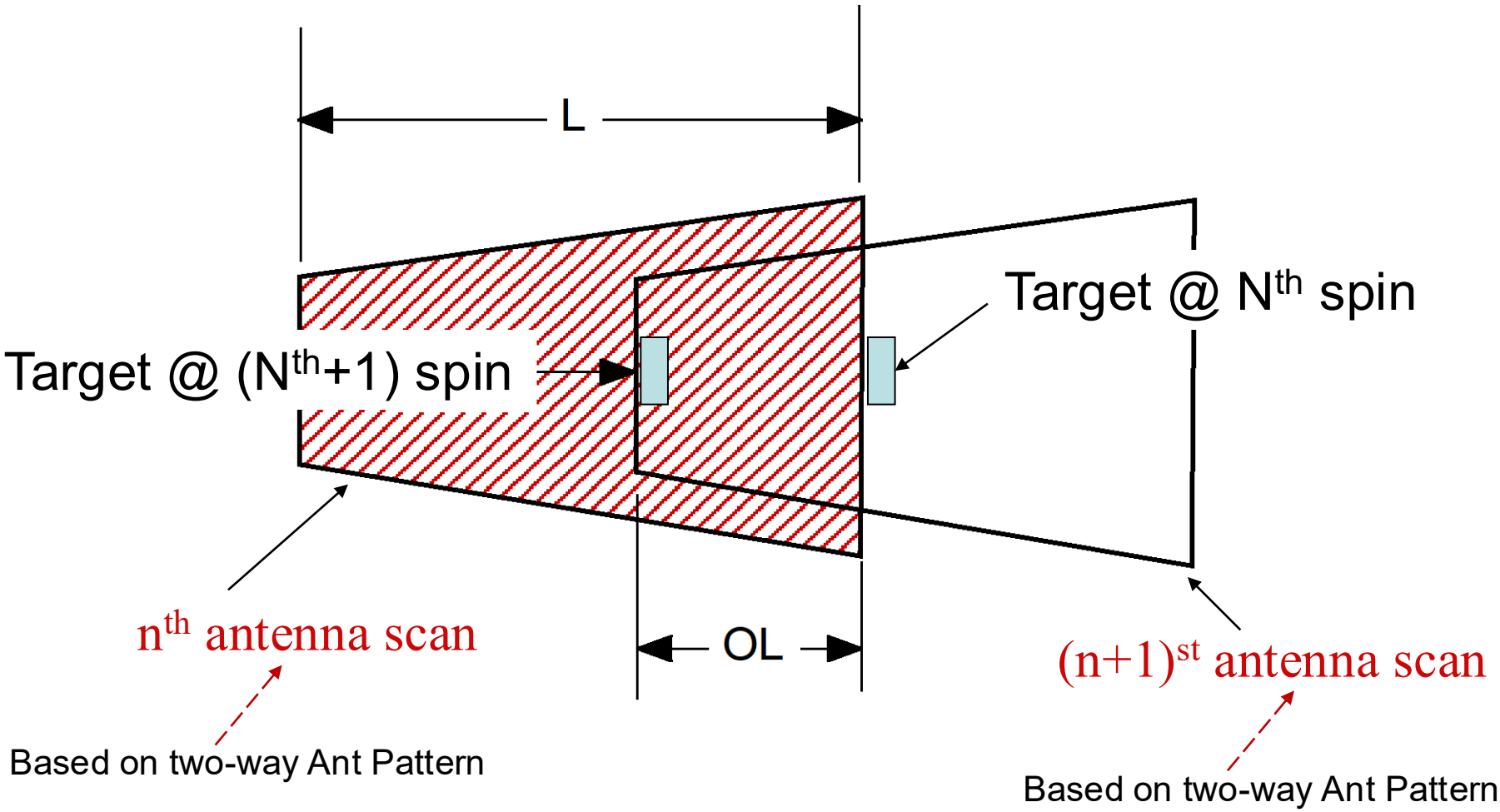


Minimum Antenna Footprint Overlap for two consecutive scans – to prevent missing target

- Target is "just within" the ant footprint for the (N^{th}+1) spin
  - if the antenna were spinning slower, the target would be missed!



Antenna Footprint Overlap for two consecutive scans - Top View

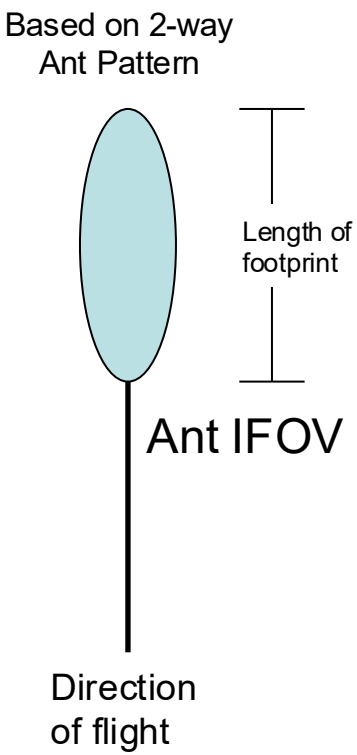


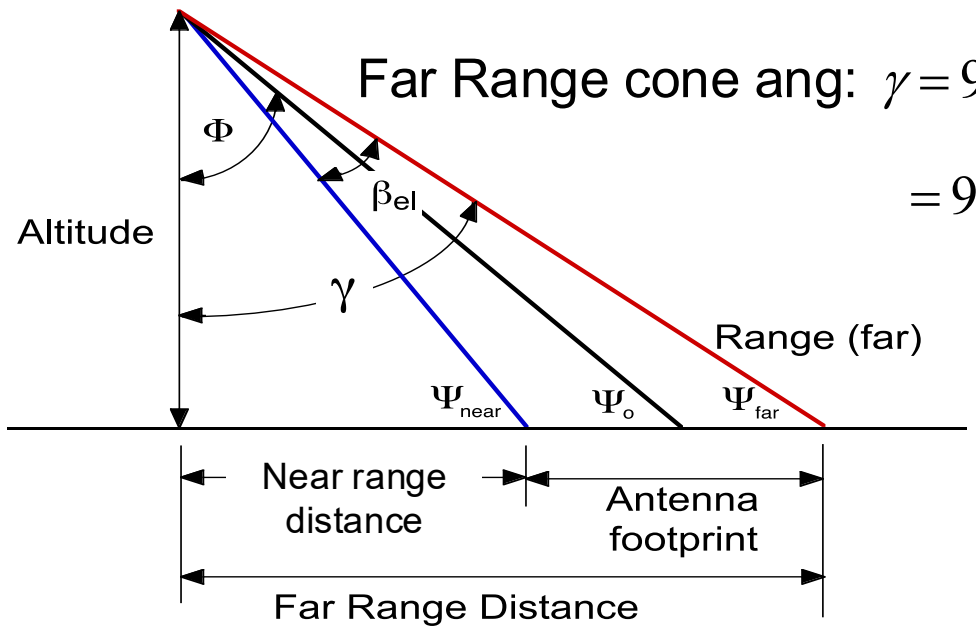
Calculate Antenna Footprint (Based on 2-way Ant Pattern)

Boresight cone ang: $\phi = 90\square - \psi_o$

Far Range cone ang: $\gamma = 90\square - \psi_o + \left(\beta_{el}/2\right)$

$= 90\square - \psi_{far}$





Calculate length of footprint:

$$length_{footprint} = dist_{Range(far)} - dist_{Range(near)}$$

$$length_{footprint} = Alt * (\tan(\gamma) - \tan(\Phi - \beta_{el}/2))$$

Minimum Spin Rate without missing a target (i.e., just detecting the target once)

$$SpinRate(rpm) = \frac{60}{Period(\sec)}$$

$$Period(\sec) = \frac{length_{footprint}}{Velocity_{relative}}$$

$$V_{rel} = V_{airplane} - V_{target} * \cos(\theta)$$

Where θ is the angle between the direction of flight of the airplane the direction of movement of the target

Minimum Spin Rate to detecting the target n-times

$$SpinRate(rpm) = \frac{60}{Period(\sec)}$$

$$Period(\sec) = \frac{length_{footprint} / n}{Velocity_{relative}}$$

$$V_{rel} = V_{airplane} - V_{target} * \cos(\theta)$$

where θ is the angle between the direction of flight of the airplane the direction of movement of the target

Example Time on Target (*TOT*)

Example Time on Target (*TOT*)

- Now that you have selected
  - An antenna boresight Ψ_o and the β_{EL} that satisfies T_{war} worst-case and the minimum spin rate to ensure <u>not missing a target</u>
  - What about the effect of *TOT*?
    - Increasing the *TOT* will maximize the pulses on target (*POT*), which will improve S/N (using non-coherent pulse integration)
      - $POT = TOT * PRF$

Constraints on PRF:

- Calculate the max PRF:
  - (assures only one pulse on the surface at a time - see Pulse Radar Geometry)

$$PRF = \frac{c}{2*(R_{far} - R_{near})}$$

Constraints on PRF:

- Calculate the max PRF:
  - (assures only one pulse on the surface at a time - see Pulse Radar Geometry)

$$PRF_{max} = \frac{1}{\left(R_{far} - R_{near}\right)/c + 2\tau}$$

- Pulses on Target:

$$POT = PRF * \left[\frac{60}{spin-rate}\right] * \left[\frac{\beta_{az}}{360}\right]$$

Where spin-rate is revolutions/min and β_{az} is Ant Az HPBW, deg

Thus: spin rate, β_{el}, β_{az}, ψ_o - all interact and *POT* must be traded against other reqmts

Example Prob of Detection (*Pd*)

Example P_d Trades

- Prob of detection *(P_d)* depends upon
  - *SNR = T/(C+N) & POT*

Example P_d Trades

- Prob of detection *(P_d)* depends upon
  - *T/(C+N) & POT*
- Clutter cross section = *sig-o * pulse IFOV area*
  - Clutter sig-0 = $f(\psi)$
  - Pulse area (δR x δAz)
    - But δAz = R* β_{az} and the target range = $f(\psi, Alt)$
    - and δR = *f(pulse width)*

Example Time on Target (*TOT*)

- Prob of detection *(P_d)* depends upon
  - *T/(C+N) & POT*
- Clutter cross section = *sig-o * pulse IFOV area*
  - Clutter sig-0 = *f(ψ)*
  - Pulse area (δR x δAz)
    - But δAz = R* β_{az} and the target range = *f(ψ, Alt)*
    - and δR = *f(pulse width)*
- Noise power depends upon the receiver bandwidth, which = 1/(pulse width)

Example P_d Trades

- Prob of detection *(P_d)* depends upon
  - *T/(C+N) & POT*
- Clutter cross section = *sig-o * pulse IFOV area*
  - Clutter sig-0 = $f(\psi)$
  - Pulse area (δR x δAz)
    - But δAz = R* β_{az} and the target range = $f(\psi, Alt)$
    - and δR = *f(pulse width)*
- Noise power depends upon the receiver bandwidth, which = 1/ (pulse width)
- Target cross section = $f(\psi)$

Example cont

- POT = PRF * TOT
  - TOT depends upon antenna spin rate
    - Which will be determined by the requirement to have sufficient overlap of antenna footprints to assure no missed targets
    - Depends upon the length of the ant footprint, which is a *f(grazing ang & ant el HPBW)* and
    - Worst case relative velocity of aircraft and target

Example-5 cont

- POT depends upon the PRF and the TOT
  - TOT depends upon antenna spin rate
    - Which will be determined by the requirement to have sufficient overlap of antenna footprints to assure no missed targets
    - Depends upon the length of the ant footprint, which is a *f(grazing ang & ant el HPBW)* and
    - Worst case relative velocity of aircraft and target
  - PRF depends upon the constraint to have only one pulse within the ant footprint at any time
    - Which depends upon the differential range (Rfar - Rnear)
      - Which depends upon grazing angle & Ant El HPBW β_{el}

Example-5 cont

- POT depends upon the PRF and the TOT
  - TOT depends upon antenna spin rate
    - Which will be determined by the requirement to have sufficient overlap of antenna footprints to assure no missed targets
    - Depends upon the length of the ant footprint, which is a *f(grazing ang & ant el HPBW)* and
    - Worst case relative velocity of aircraft and target
  - PRF depends upon the constraint to have only one pulse within the ant footprint at any time
    - Which depends upon the differential range (Rfar - Rnear)
      - Which depends upon grazing angle & Ant El HPBW β_{el}

Based on two-way Ant Pattern

Example cont

- POT depends upon the PRF and the TOT
  - TOT depends upon antenna spin rate
    - Which will be determined by the requirement to have sufficient overlap of antenna footprints to assure no missed targets
    - Depends upon the length of the ant footprint, which is a *f(grazing ang & ant el HPBW)* and
    - Worst case relative velocity of aircraft and target
  - PRF depends upon the constraint to have only one pulse within the ant footprint at any time
    - Which depends upon the differential range (Rfar - Rnear)
      - Which depends upon grazing angle & ant el HPBW β_{el}
  - Also PRF must satisfy the Nyquist criteria when measuring Doppler
    - depends upon target/AC radial component of velocity

Example Prob of Target Detection

Example Prob of Target Detection

- The intent of radar is to detect "point" targets in the presence of thermal and other forms of noise
- Unintentional echo's (Clutter) can result from illuminating distributed "surface" targets
  - Earth's surface (land, ice & ocean)

Sect 2.13: Surface Clutter (σ_{sc}) Radar Cross Section

- The "instantaneous illuminated area" (IFOV) by the radar pulse determines the clutter power

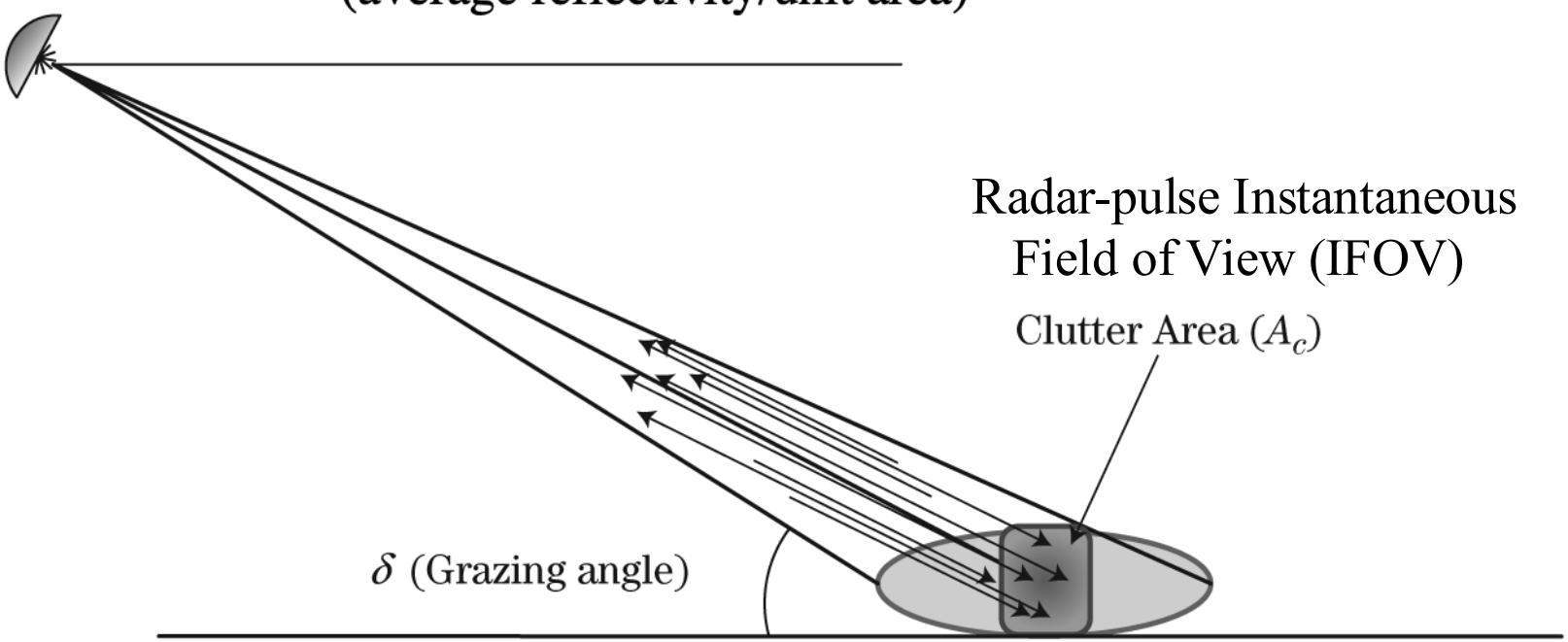
Based on two-way Ant Pattern

$$\sigma_{sc} = IFOV_{pulse} * \sigma^o$$

– where $IFOV_{pulse}$ is the pulse-illumination on the surface = $\delta R * \delta Az$

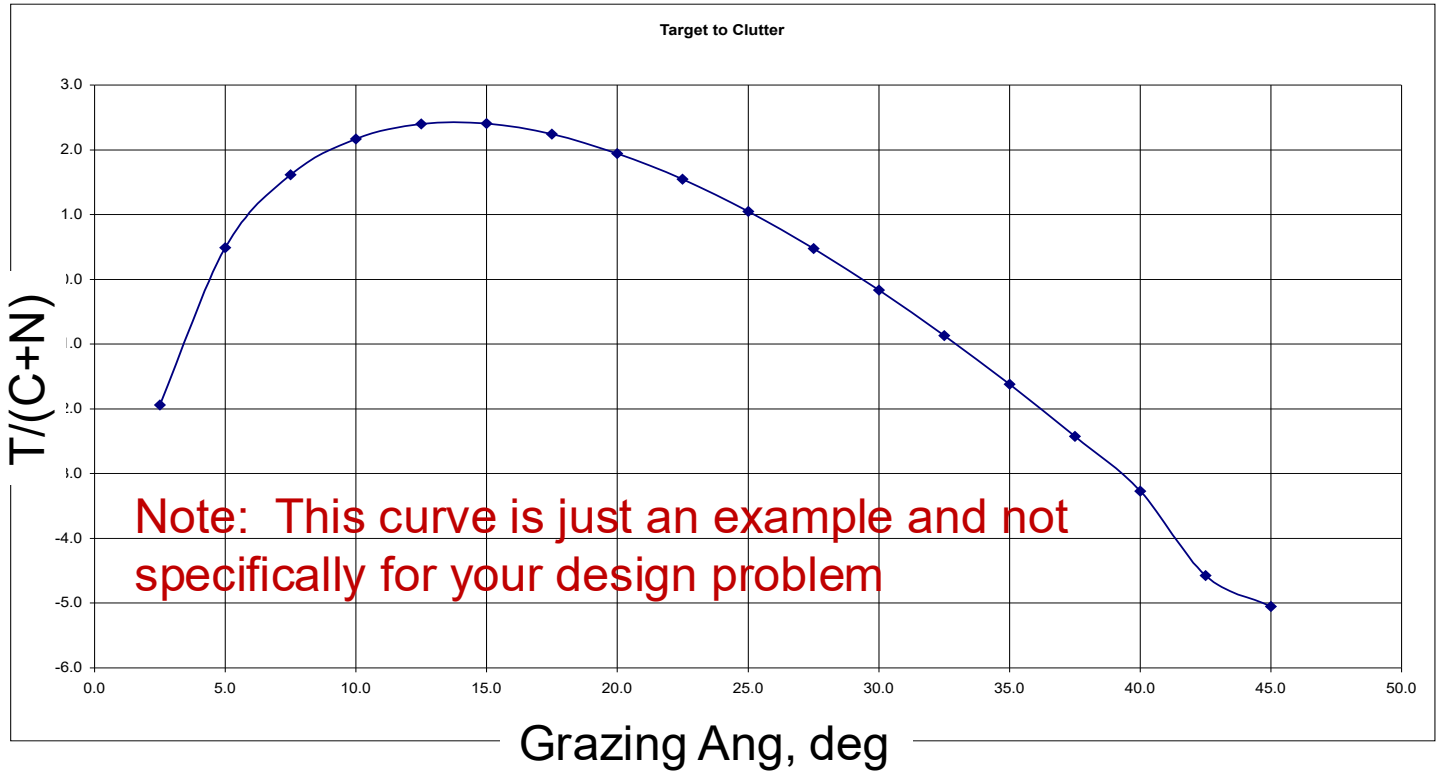
σ^o is the normalized radar cross section

= (average reflectivity/unit area)



Example cont

- So, a critical decision is where to point the antenna boresight i.e., what grazing angle @ boresight?
  - You should trade pointing at high grazing angles (near the aircraft) versus low grazing angles (far from the aircraft)
    - Try to optimize T/(C+N) by varying other radar parameters at selected ant boresight grazing angles e.g., see below:



Have your radar design checked by comparison with mine!

- I will meet with groups individually to check your calculations
  - To perform a check, I will require that you answer the Q? at the end of the spec concerning radar parameters and the resulting calculations

Have your radar design checked by comparison with mine!

- I will meet with groups individually to check your calculations
  - To perform a check, I will require that you answer the Q? at the end of the spec concerning radar parameters and the resulting calculations
  - You do NOT have to complete your design to have it checked

Have your radar design checked by comparison with mine!

- I will meet with groups individually to check your calculations
  - To perform a check, I will require that you answer the Q? at the end of the spec concerning radar parameters and the resulting calculations
  - You do NOT have to complete your design to have it checked
  - There are no negative consequences for taking advantage of this “confidential” consulting service

Have your radar design checked by comparison with mine!

- I will meet with groups individually to check your calculations
  - To perform a check, I will require that you answer the Q? at the end of the spec concerning radar parameters and the resulting calculations
  - You do NOT have to complete your design to have it checked
  - There are no negative consequences for taking advantage of this "confidential" consulting service
  - Total design time is estimated to be **30 – 40 hrs**
- Good luck and start early!

Example: Evaluation Report

Evaluation

Specs

Proposed Aircraft Alt	-	m
Ground speed	115	km/hr
Ground speed	31.9	m/s
Antenna (RMP) - max	180.0	rpm
Proposed Ant RPM	0	rpm
RPM goal	30	rpm
# scans to track target (min)	3	scans
Proposed # scans to track	0	scans
# scans target track goal	10	scans
MaxAnt Elev diameter	1.2	m
Proposed Ant Elev diameter	0.000	m
Min. Ant.dia El goal, m	0.10	m
Max Ant Az diameter	1.2	m
Proposed Ant Az diameter	0.00	m
Min ant dia az goal, m	0.10	m
Proposed Boresight Grazing Ang	0.0	deg
Min. grazing angle	2.0	deg
Proposed Az +/- from fore-look	0.0	deg
Proposed Radar Freq	0.000E+00	Hz
Xmtr pwr (max)	1500	W
Proposed Transmit Pwr	0.00	W
Xmtr pwr (goal)	10	W
Proposed Transmit pulse width	0.0000	microsec
Xmtr duty cycle	10.0	%
Proposed PRF	-	Hz
PRF max	500	KHz
Proposed PCR	0.00	
System noise temp - f(dfreq)	700.0	K
Atmos propagation loss - f(freq)	-0.400	dB/km
Target prob of detection - min	0.55	
Target prob of det (2of4) - goal	0.95	
Proposed <Tfa>	0.00	sec
<Tfa> = #*Ts, sec	10	
Target cross section	Variable	m^2
Target max dimension	6.0	m
Clutter sigma-0	variable	m^2
Max Target Velocity	200.0	km/hr
	55.56	m/s
Min target radial velocity	5.0	m/s
Min warning time	6.0	s
Warning time (goal)	60	s
Doppler Meas error*	700	Hz
Doppler Prec goal	50	Hz
Swath width, min	10	km
Proposed swath	#DIV/0!	
Swath goal, km	50	km

Evaluation

Your Inputs

Radar Design Parameters Choices		
Radar Freq		Hz
UAV Altitude		km
Ant.boresight grazing ang		deg
Ant dia- elevation		meter
Ant dia- azimuth		meter
Ant spin rate		rpm
Antenna Azimuth +/- from fore-look		deg
Xmtr Pwr		W
Xmt pulse width		microsec
PRF		Hz
<Tfa>		sec
Design capability: # scans on target = N		integer
PCR		
Detection to Doppler (Time/scan)*		ratio

Proposed Aircraft Alt	-	m
Ground speed	115	km/hr
Ground speed	31.9	m/s
Antenna (RMP) - max	180.0	rpm
Proposed Ant RPM	0	rpm
RPM goal	30	rpm
# scans to track target (min)	3	scans
Proposed # scans to track	0	scans
# scans target track goal	10	scans
MaxAnt Elev diameter	1.2	m
Proposed Ant Elev diameter	0.000	m
Min. Ant.dia El goal, m	0.10	m
Max Ant Az diameter	1.2	m
Proposed Ant Az diameter	0.00	m
Min ant dia az goal, m	0.10	m
Proposed Boresight Grazing Ang	0.0	deg
Min. grazing angle	2.0	deg
Proposed Az +/- from fore-look	0.0	deg
Proposed Radar Freq	0.000E+00	Hz
Xmtr pwr (max)	1500	W
Proposed Transmit Pwr	0.00	W
Xmtr pwr (goal)	10	W
Proposed Transmit pulse width	0.0000	microsec
Xmtr duty cycle	10.0	%
Proposed PRF	-	Hz
PRF max	500	KHz
Proposed PCR	0.00	
System noise temp - f(dfreq)	700.0	K
Atmos propagation loss - f(freq)	-0.400	dB/km
Target prob of detection - min	0.55	
Target prob of det (2of4) - goal	0.95	
Proposed <Tfa>	0.00	sec
<Tfa> = #*Ts, sec	10	
Target cross section	Variable	m^2
Target max dimension	6.0	m
Clutter sigma-0	variable	m^2
Max Target Velocity	200.0	km/hr
	55.56	m/s
Min target radial velocity	5.0	m/s
Min warning time	6.0	s
Warning time (goal)	60	s
Doppler Meas error*	700	Hz
Doppler Prec goal	50	Hz
Swath width, min	10	km
Proposed swath	#DIV/0!	
Swath goal, km	50	km

Evaluation: Geometry Test Case Antenna HPBW

Radar Design Parameters Choices		
Radar Freq	3.10E+10	Hz
UAV Altitude	10	km
Ant.boresight grazing ang	20.0	deg
Ant dia- elevation	1.000	meter
Ant dia- azimuth	0.50	meter
Ant spin rate		rpm
Antenna Azimuth +/- from fore-look		deg
Xmtr Pwr		W
Xmt pulse width	1.0000	microsec
PRF		Hz
<Tfa>		sec
Design capability: # scans on target = N		integer
PCR	1.00	
Detection to Doppler (Time/scan)*		ratio

Beta_El , deg	0.67	
Beta_Az, deg	1.33	
Radar X-factor, dB		#NUM!
Swath, km	0.00	

Evaluation: Geometry Test Case

Radar Design Parameters Choices		
Radar Freq	3.10E+10	Hz
UAV Altitude	10	km
Ant.boresight grazing ang	20.0	deg
Ant dia- elevation	1.000	meter
Ant dia- azimuth	0.50	meter
Ant spin rate		rpm
Antenna Azimuth +/- from fore-look		deg
Xmtr Pwr		W
Xmt pulse width	1.0000	microsec
PRF		Hz
<Tfa>		sec
Design capability: # scans on target = N		integer
PCR	1.00	
Detection to Doppler (Time/scan)*		ratio

Diagnostics:	linear	dB	Linear	dB	Linear	dB
Geometry	near		bore		Far	
Grazing angle, deg	20.33		20.00		19.67	
Beta_El , deg	0.67					
Beta_Az, deg	1.33					
Radar X-factor, dB		#NUM!				
Swath, km	0.00					

$$\Psi_{near} = \Psi_o + (\beta_{EL})/2$$

$$\Psi_{far} = \Psi_o - (\beta_{EL})/2$$

Evaluation: Geometry Test Case

Radar Design Parameters Choices		
Radar Freq	3.10E+10	Hz
UAV Altitude	10	km
Ant.boresight grazing ang	20.0	deg
Ant dia- elevation	1.000	meter
Ant dia- azimuth	0.50	meter
Ant spin rate		rpm
Antenna Azimuth +/- from fore-look		deg
Xmtr Pwr		W
Xmt pulse width	1.0000	microsec
PRF		Hz
<Tfa>		sec
Design capability: # scans on target = N		integer
PCR	1.00	
Detection to Doppler (Time/scan)*		ratio

Diagnostics:	linear	dB	Linear	dB	Linear	dB
Geometry	near		bore		Far	
Range, m	28779.4		29238.0		29712.5	
Grazing angle, deg	20.33		20.00		19.67	
Beta_El , deg	0.67					
Beta_Az, deg	1.33					
Radar X-factor, dB		#NUM!				
Swath, km	0.00					

Evaluation: Geometry Test Case

Radar Design Parameters Choices		
Radar Freq	3.10E+10	Hz
UAV Altitude	10	km
Ant.boresight grazing ang	20.0	deg
Ant dia- elevation	1.000	meter
Ant dia- azimuth	0.50	meter
Ant spin rate		rpm
Antenna Azimuth +/- from fore-look		deg
Xmtr Pwr		W
Xmt pulse width	1.0000	microsec
PRF		Hz
<Tfa>		sec
Design capability: # scans on target = N		integer
PCR	1.00	
Detection to Doppler (Time/scan)*		ratio

Diagnostics:	linear	dB	Linear	dB	Linear	dB
Geometry	near		bore		Far	
Range, m	28779.4		29238.0		29712.5	
Grazing angle, deg	20.33		20.00		19.67	
Pulse Range resolution = dR, m	159.967		159.627		159.293	

$$dR = (c\tau/2)/\cos(\text{grazing})$$

Beta_El , deg	0.67	
Beta_Az, deg	1.33	
Radar X-factor, dB		#NUM!
Swath, km	0.00	

Evaluation: Geometry Test Case

Radar Design Parameters Choices		
Radar Freq	3.10E+10	Hz
UAV Altitude	10	km
Ant.boresight grazing ang	20.0	deg
Ant dia- elevation	1.000	meter
Ant dia- azimuth	0.50	meter
Ant spin rate		rpm
Antenna Azimuth +/- from fore-look		deg
Xmtr Pwr		W
Xmt pulse width	1.0000	microsec
PRF		Hz
<Tfa>		sec
Design capability: # scans on target = N		integer
PCR	1.00	
Detection to Doppler (Time/scan)*		ratio

Diagnostics:	linear	dB	Linear	dB	Linear	dB
Geometry	near		bore		Far	
Range, m	28779.4		29238.0		29712.5	
Grazing angle, deg	20.33		20.00		19.67	
Pulse Range resolution = dR, m	159.967		159.627		159.293	
Pulse Azimuth resolution= dAz, m	668.43		679.08		690.10	

$$dAz_{near} = (R_{near})*(\beta_{az_radians})$$

Beta_Az, deg	1.33	
Radar X-factor, dB		#NUM!
Swath, km	0.00	

Evaluation: Geometry Test Case

Radar Design Parameters Choices		
Radar Freq	3.10E+10	Hz
UAV Altitude	10	km
Ant.boresight grazing ang	20.0	deg
Ant dia- elevation	1.000	meter
Ant dia- azimuth	0.50	meter
Ant spin rate		rpm
Antenna Azimuth +/- from fore-look		deg
Xmtr Pwr		W
Xmt pulse width	1.0000	microsec
PRF		Hz
<Tfa>		sec
Design capability: # scans on target = N		integer
PCR	1.00	
Detection to Doppler (Time/scan)*		ratio

Diagnostics:	linear	dB	Linear	dB	Linear	dB
Geometry	near		bore		Far	
Range, m	28779.4		29238.0		29712.5	
Grazing angle, deg	20.33		20.00		19.67	
Pulse Range resolution = dR, m	159.967		159.627		159.293	
Pulse Azimuth resolution= dAz, m	668.43		679.08		690.10	
Pulse IFOV, m^2	106926.24		108398.82		109927.57	

Pulse IFOV = dR*dAz

Beta_El , deg	0.67	
Beta_Az, deg	1.33	
Radar X-factor, dB		#NUM!
Swath, km	0.00	

Evaluation: Geometry Test Case

Radar Design Parameters Choices		
Radar Freq	3.10E+10	Hz
UAV Altitude	10	km
Ant.boresight grazing ang	20.0	deg
Ant dia- elevation	1.000	meter
Ant dia- azimuth	0.50	meter
Ant spin rate		rpm
Antenna Azimuth +/- from fore-look		deg
Xmtr Pwr		W
Xmt pulse width	1.0000	microsec
PRF		Hz
<Tfa>		sec
Design capability: # scans on target = N		integer
PCR	1.00	
Detection to Doppler (Time/scan)*		ratio

Diagnostics:	linear	dB	Linear	dB	Linear	dB
Geometry	near		bore		Far	
Range, m	28779.4		29238.0		29712.5	
Grazing angle, deg	20.33		20.00		19.67	
Pulse Range resolution = dR, m	159.967		159.627		159.293	
Pulse Azimuth resolution= dAz, m	668.43		679.08		690.10	
Pulse IFOV, m^2	106926.24		108398.82		109927.57	
Target Cross section, m^2	3.87	5.86	3.86	5.86	3.85	5.85
Beta_Az, deg	1.33					
Radar X-factor, dB		#NUM!				
Swath, km	0.00					

$\sigma_T = f(\text{grazing ang})$, dB

Evaluation: Geometry Test Case

Radar Design Parameters Choices		
Radar Freq	3.10E+10	Hz
UAV Altitude	10	km
Ant.boresight grazing ang	20.0	deg
Ant dia- elevation	1.000	meter
Ant dia- azimuth	0.50	meter
Ant spin rate		rpm
Antenna Azimuth +/- from fore-look		deg
Xmtr Pwr		W
Xmt pulse width	1.0000	microsec
PRF		Hz
<Tfa>		sec
Design capability: # scans on target = N		integer
PCR	1.00	
Detection to Doppler (Time/scan)*		ratio

Diagnostics:	linear	dB	Linear	dB	Linear	dB
Geometry	near		bore		Far	
Range, m	28779.4		29238.0		29712.5	
Grazing angle, deg	20.33		20.00		19.67	
Pulse Range resolution = dR, m	159.967		159.627		159.293	
Pulse Azimuth resolution= dAz, m	668.43		679.08		690.10	
Pulse IFOV, m^2	106926.24		108398.82		109927.57	
Boresight Target Cross section, m^2	3.87	5.86	3.86	5.86	3.85	5.85
Clutter cross section m^2	795.605	29.01	780.670	28.92	765.380	-1.80
Radar X-factor, dB		#NUM!				
Swath, km	0.00					

$\sigma_C = (\text{Pulse_IFOV})^* \sigma_o$

$\sigma_o = f(\text{grazing ang})$, dB

Evaluation: Geometry Test Case

Radar Design Parameters Choices		
Radar Freq	3.10E+10	Hz
UAV Altitude	10	km
Ant.boresight grazing ang	20.0	deg
Ant dia- elevation	1.000	meter
Ant dia- azimuth	0.50	meter
Ant spin rate		rpm
Antenna Azimuth +/- from fore-look		deg
Xmtr Pwr		W
Xmt pulse width	1.0000	microsec
PRF		Hz
<Tfa>		sec
Design capability: # scans on target = N		integer
PCR	1.00	
Detection to Doppler (Time/scan)*		ratio

Diagnostics:	linear	dB	Linear	dB	Linear	dB
Geometry	near		bore		Far	
Range, m	28779.4		29238.0		29712.5	
Grazing angle, deg	20.33		20.00		19.67	
Pulse Range resolution = dR, m	159.967		159.627		159.293	
Pulse Azimuth resolution= dAz, m	668.43		679.08		690.10	
Pulse IFOV, m^2	106926.24		108398.82		109927.57	
Boresight Target Cross section, m^2	3.87	5.86	3.86	5.86	3.85	5.85
Clutter cross section m^2	795.605	29.01	780.670	28.92	765.380	-1.80
Ant footprint length. M	993.01					
footprint overlap scan to scan, %	#DIV/0!					
Beta_El , deg	0.67					
Beta_Az, deg	1.33					
Radar X-factor, dB		#NUM!				
Swath, km	0.00					

New Case

Radar Freq	3.70E+10	Hz
UAV Altitude	10	km
Ant.boresight grazing ang	50.0	deg
Ant dia- elevation	0.150	meter
Ant dia- azimuth	1.00	meter
Ant spin rate	10	rpm
Antenna Azimuth +/- from fore-look	60.0	deg
Xmtr Pwr	50.00	W
Xmt pulse width	0.0500	microsec
PRF	250,000	Hz
<Tfa>	60.00	sec
Design capability: # scans on target = N	3	integer
PCR	1.00	
Detection to Doppler (Time/scan)*		ratio

Evaluation

Non-compliant Specs

Outputs			Spec	Meets?	Score
Swath width, km	15.52		10.0	yes	1
Xmt pulse width, sec	5.0000E-08			yes	1
Antenna Az dis, m	1.00		1	yes	1
Antenna El dia, m	0.15	≤	1	yes	1
Min grazing ang	48.1	≤	2	yes	1
max allowable PRF for delta range	210,775.16				
Design selected PRF OK? Hz	250,000.00		500,000	NO!	0

Evaluation

Non-compliant Specs

Outputs			Spec	Meets?	Score
Swath width, km	15.52		10.0	yes	1
Xmt pulse width, sec	5.0000E-08			yes	1
Antenna Az dis, m	1.00		1	yes	1
Antenna El dia, m	0.15	≤	1	yes	1
Min grazing ang	48.1	≤	2	yes	1
max allowable PRF for delta range	210,775.16				
Design selected PRF OK? Hz	250,000.00		500,000	NO!	0
<Tfa>, #spin period	10.00	≥	10	yes	1
Ant. Spin rate, rpm (input)	10.00	≤	180	yes	1
Design capability # scans on target = N	50.0	≥	3.0	yes	1
Min spin rate = no target missed xN scans?	14.23				
Does ant spin rate satisfy "no missed target" spec?	10.00		14.23	NO!	0

Evaluation

Non-compliant Specs

Outputs			Spec	Meets?	Score
Swath width, km	15.52		10.0	yes	1
Xmt pulse width, sec	5.0000E-08			yes	1
Antenna Az dis, m	1.00		1	yes	1
Antenna El dia, m	0.15	≤	1	yes	1
Min grazing ang	48.1	≤	2	yes	1
max allowable PRF for delta range	210,775.16				
Design selected PRF OK? Hz	250,000.00		500,000	NO!	0
<Tfa>, #spin period	10.00	≥	10	yes	1
Ant. Spin rate, rpm (input)	10.00	≤	180	yes	1
Design capability # scans on target = N	50.0	≥	3.0	yes	1
Min spin rate = no target missed xN scans?	14.23				
Does ant spin rate satisfy "no missed target" spec?	10.00		14.23	NO!	0
min warning time - worst case @ edge of scan (sec)	44.87	≥	6	yes	1
Xmt Pwr (new 2-15-25)	50.00		1.50E+03	yes	1
Xmtr duty cycle	0.01250	≤	0.1005	yes	1
Range Resolution @ boresight, m	11.24	≥	6	yes	1
Delta-Az @ bore, m	127.0126	≥	6	yes	1
Prob of detection @ Rnear	0.388		0.55	NO!	0
Prob of detection @ Bore	0.641		0.55	yes	1
Prob of detection @ Rfar	0.801		0.55	yes	1

Pd single scan is no longer specified

New spec is Pd(2 of N) ≥ 0.8

Evaluation

Non-compliant Specs

Outputs			Spec	Meets?	Score
Swath width, km	15.52		10.0	yes	1
Xmt pulse width, sec	5.0000E-08			yes	1
Antenna Az dis, m	1.00		1	yes	1
Antenna El dia, m	0.15	≤	1	yes	1
Min grazing ang	48.1	≤	2	yes	1
max allowable PRF for delta range	210,775.16				
Design selected PRF OK? Hz	250,000.00		500,000	NO!	0
<Tfa>, #spin period	10.00	≥	10	yes	1
Ant. Spin rate, rpm (input)	10.00	≤	180	yes	1
Design capability # scans on target = N	50.0	≥	3.0	yes	1
Min spin rate = no target missed xN scans?	14.23				
Does ant spin rate satisfy "no missed target" spec?	10.00		14.23	NO!	0
min warning time - worst case @ edge of scan (sec)	44.87	≥	6	yes	1
Xmt Pwr (new 2-15-25)	50.00		1.50E+03	yes	1
Xmtr duty cycle	0.01250	≤	0.1005	yes	1
Range Resolution @ boresight, m	11.24	≥	6	yes	1
Delta-Az @ bore, m	127.0126	≥	6	yes	1
Prob of detection @ Rnear	0.388		0.55	NO!	0
Prob of detection @ Bore	0.641		0.55	yes	1
Prob of detection @ Rfar	0.801		0.55	yes	1
Doppler TOT (for selected # scans)	2.79E-02				
Nyquist PRF for doppler meas, Hz	28,804.68				
Max Doppler, Hz	14,402.34				
PRF satisfied Nyquist?	250,000			yes	1
Doppler meas precision worst case			700	yes	1
Design compliant score					0%

Modified input

Radar Design Parameters Choices		
Radar Freq	3.70E+10	Hz
UAV Altitude	10	km
Ant.boresight grazing ang	50.0	deg
Ant dia- elevation	0.150	meter
Ant dia- azimuth	1.00	meter
Ant spin rate	15	rpm
Antenna Azimuth +/- from fore-look	60.0	deg
Xmtr Pwr	1500.00	W
Xmt pulse width	0.0400	microsec
PRF	200,000	Hz
<Tfa>	60.00	sec
Design capability: # scans on target = N	3	integer
PCR	1.00	
Detection to Doppler (Time/scan)*		ratio

Outputs			Spec	Meets?	Score
Swath width, km	15.52		10.0	yes	1
Xmt pulse width, sec	4.0000E-08			yes	1
Antenna Az dis, m	1.00		1	yes	1
Antenna El dia, m	0.15	≤	1	yes	1
Min grazing ang	48.1	≤	2	yes	1
max allowable PRF for delta range	210,775.16				
Design selected PRF OK? Hz	200,000.00		500,000	yes	1
<Tfa>, #spin period	15.00	≥	10	yes	1
Ant. Spin rate, rpm (input)	15.00	≤	180	yes	1
Design capability # scans on target = N	50.0	≥	3.0	yes	1
Min spin rate = no target missed xN scans?	14.23				
Does ant spin rate satisfy "no missed target" spec?	15.00		14.23	yes	1
min warning time - worst case @ edge of scan (sec)	44.87	≥	6	yes	1
Xmt Pwr (new 2-15-25)	1500.00		1.50E+03	yes	1
Xmtr duty cycle	0.00800	≤	0.1005	yes	1
Range Resolution @ boresight, m	8.99	≥	6	yes	1
Delta-Az @ bore, m	127.0126	≥	6	yes	1
Prob of detection @ Rnear	0.617		0.55	yes	1
Prob of detection @ Bore	0.915		0.55	yes	1
Prob of detection @ Rfar	0.988		0.55	yes	1
Doppler TOT (for selected # scans)	1.86E-02				
Nyquist PRF for doppler meas, Hz	28,804.68				
Max Doppler, Hz	14,402.34				
PRF satisfied Nyquist?	200,000			yes	1
Doppler meas precision worst case	[redacted]		700	yes	1
Design compliant score					100%
Non-compliant design score = {(16 - #"NO's)/16} 50%					NA

Fully compliant Perf

Goals

Goals			Goal	Meets?	% of Goal
Ant El dia	FALSE		0.10	NO!	95%
Ant Az dia	FALSE		0.10	NO!	18%
Target tracking, # spins	FALSE		10	NO!	0%
Ant Spin rate	TRUE		30	yes	100.0%
Warning time	FALSE		60	NO!	72%
Prob of detection Pd(2 of 4)	TRUE		0.95	NO!	100%
Doppler precision goal	FALSE		50.0	yes	100%
Transmit power	FALSE		10.0	NO!	0%
Swath width	FALSE		5.00E+04	NO!	13.80%
Total goals score					55.49%

Good Luck and Start Early!